

RotorLab — Final Results: MRF vs Sliding Mesh vs Benchmark

Capstone comparison for the Rushton turbine stirred-tank benchmark ($T=0.300\text{ m}$, $D=0.100\text{ m}$, $N=5\text{ rev/s}$, water, 6-blade disc turbine, 4 wall baffles). Companion to the Build & Decision Log, MRF Results Report, and Sliding Mesh Setup Log.

Executive Summary

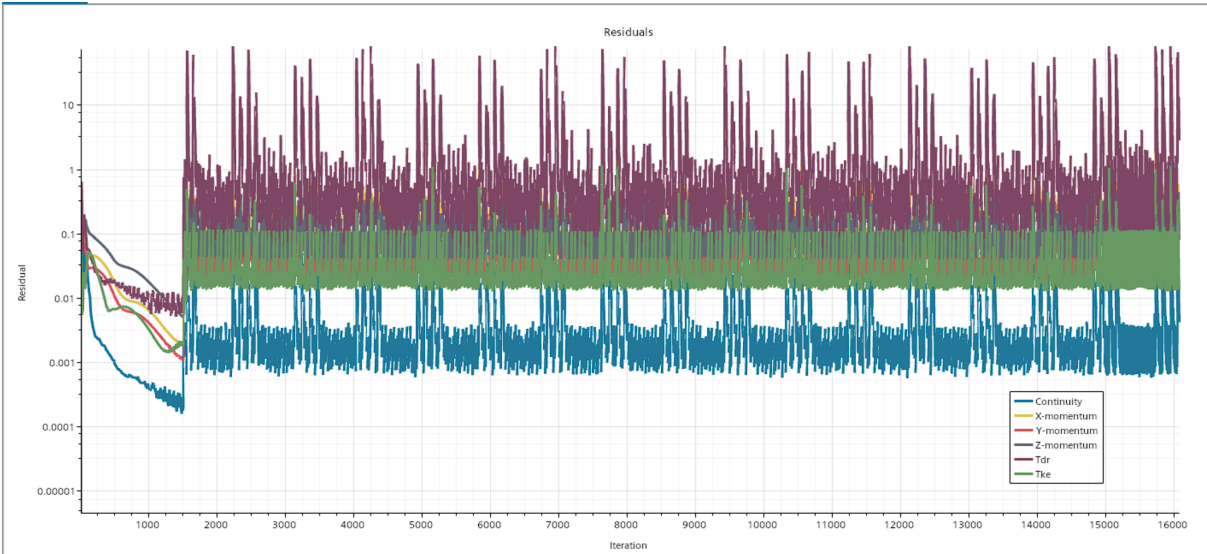
Both the steady MRF and transient sliding-mesh methods were run to completion on identical geometry and mesh. They agree closely with **each other** (Po 10.49 vs. 10.67, $\sim 1.7\%$ apart) but both disagree with the published benchmark (Po ≈ 5.0) by essentially the same $\sim 2.1\times$ factor. This is the key finding of the project: going from steady to fully transient did **not** close the gap to the benchmark, which means the original working hypothesis — that steady MRF's inability to resolve blade-passage unsteadiness was the primary cause of the discrepancy — is not well supported by the data. A factor common to both simulations (most likely the Realizable K-Epsilon turbulence model, documented in the literature to over-predict power draw for this geometry class) is the more likely explanation. Sliding mesh did deliver real, independent value: it resolved a genuine periodic torque signal MRF cannot produce at all.

1. Results Comparison

Metric	Steady MRF	Sliding Mesh	Published Benchmark
Torque magnitude	0.41666 N·m	0.42362 N·m (period-averaged)	—
Power Number (Po)	10.49	10.67	≈ 5.0
Ratio vs. benchmark	$\approx 2.10\times$	$\approx 2.13\times$	1.00 $\times$
Ratio vs. each other	—	1.017 $\times$ MRF (1.7% apart)	—
Convergence basis	1500 steady iterations, flat torque	Time-averaged over a periodic window ($\sim 0.736\text{--}0.810\text{ s}$, 130 timesteps)	—

Sliding mesh torque was time-averaged from Monitor data covering Time $\approx 0.736\text{--}0.810\text{ s}$ (near the end of an 0.81 s / ~ 8 -revolution run), recovered from screenshots after the primary CSV export was found to be corrupted (see Section 5). This window is slightly short of one full $\sim 0.1\text{ s}$ fundamental period (see Section 3) rather than an exact whole-period average, so treat the sliding-mesh Po as a solid working estimate, not a laboratory-grade final digit.

2. Convergence Evidence

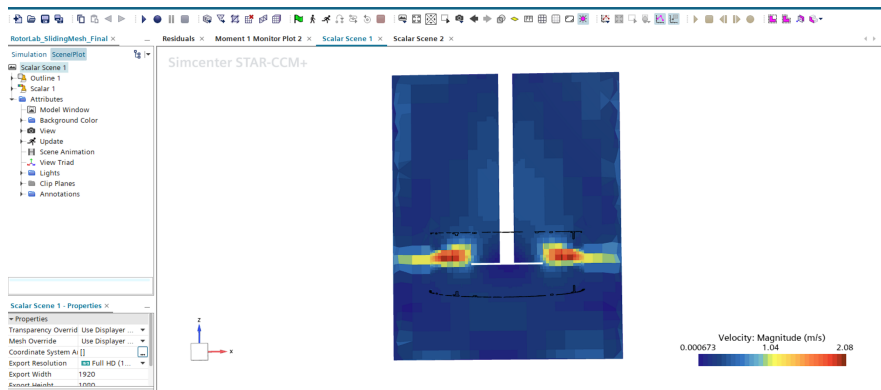


Residuals across the full run (Iteration 0-16000+). The first ~ 1500 iterations are the steady MRF phase converging normally. After the switch to sliding mesh, residuals stop trending toward zero and instead settle into a repeating banded pattern — the correct signature of a solution that has locked onto a genuine periodic state, not a sign of non-convergence (see Sliding Mesh Setup Log, Section 10, for the full reasoning).

3. Periodicity Finding

The sliding-mesh torque signal settled into a clear repeating pattern with a period of ≈ 0.1 s — **half a revolution**, not a full revolution (0.2 s). This matches the geometry exactly: with 6 blades and 4 baffles, $\text{gcd}(6,4)=2$, so the combined rotor-stator spatial pattern only needs to rotate 180° before it repeats. This is a genuine, physically grounded result that steady MRF is structurally incapable of producing, since MRF freezes the rotor at one angular position for the entire solve.

4. Flow Field

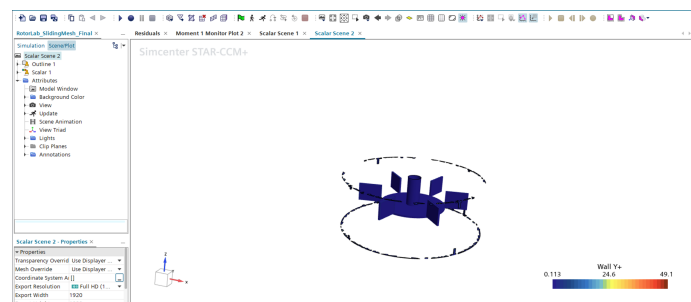


Velocity magnitude (Lab Reference Frame) on a plane section through the tank, sliding-mesh case. The classic Rushton signature is visible: a high-velocity radial discharge jet (red/orange, up to 2.08 m/s) firing outward from the impeller and splitting into upper and lower circulation loops through the bulk tank — confirming both `StationaryFluid` and `RotatingFluid` are fully coupled and the whole-tank mixing pattern is captured, not just the rotating zone.

5. Data Quality Notes

- **Solution History (animation) compromised, not used.** Repeated stop/resume cycles (required because the remote session would otherwise close unsaved) caused the .simh file's internal snapshot numbering to desync, producing recurring HDF5 "name already exists" write failures. Confirmed non-fatal to the core solve — Time-Step and Physical Time were independently verified to stay exactly consistent with Δt across every stop/resume boundary. Solution History was disabled partway through; no flow-field animation was produced for this run.
- **One CSV export was corrupted** (every row showed an identical Time value with cycling Moment values resembling inner-iteration data, not one-value-per-timestep data). Discarded. The Section 1 average instead uses Time-Step-triggered Monitor data recovered from screenshots, transcribed and computed programmatically rather than read off a plot.
- **Solver-level "Breaking out of time step iteration / Time solution is in an inconsistent state" messages** appeared at every stop/resume boundary. Confirmed benign: each was immediately followed by "resuming time step" and a clean redo of the interrupted step, with no discontinuity in the following data.

6. Near-Wall Mesh Check



Wall Y^+ on ImpellerWall, sliding-mesh case: range 0.113-49.1, visually dominated by low values across most of the surface (same pattern as the MRF check) — consistent, not a meaningful contributor to the benchmark gap.

7. Discussion: Why the Benchmark Gap Persists

The working theory from the MRF phase was that steady MRF's inability to resolve blade-baffle interaction was the primary driver of the $\sim 2.1\times$ over-prediction. Sliding mesh was the direct test of that theory: if true, resolving real blade-passage unsteadiness should have pulled Po substantially toward the benchmark. It did not — sliding mesh landed at $Po=10.67$, essentially the same as MRF's 10.49, not closer to 5.0. That result reframes the explanation. The most likely remaining driver is the turbulence model: Realizable K-Epsilon is documented in the literature to over-predict power draw for Rushton impellers specifically, independent of whether the simulation is steady or transient — since both runs share the same turbulence closure,

both would inherit the same bias. Mesh resolution was checked and is not a strong candidate (Section 6, and the equivalent MRF check). A mismatch between this setup's exact conditions and the published correlation's assumptions (Reynolds number regime, clearance ratio, exact impeller proportions) is also possible and not ruled out here.

8. Conclusions & Recommended Next Steps

The project successfully built and validated two independent numerical approaches to the same benchmark, with internally consistent, mutually agreeing results — a legitimate outcome even though neither matches the published Po. The sliding-mesh run additionally recovered a real, physically grounded periodic flow structure (Section 3) that is only accessible via transient methods. For closing the benchmark gap specifically, the next highest-value step would be a turbulence-model sensitivity check (e.g. SST k- ω or a scale-resolving method) rather than further mesh refinement or additional transient runtime, given the evidence in Sections 5-7 that both of those were already checked and are unlikely primary causes.